LEXIGRID

Email Configuration Administration

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Shahryar Jahangir, Aniza Inc.

Procedures, examples, verification and operational limits. Complete commissioning on your deployment before serving customer traffic. Sample values are fictional; original keys and data must be preserved.

Contents

Email configuration operations 3

Global configuration and roles . . . 3
Configuration procedure . . . 3
Verification without accidental send . . . 3
Error recovery . . . 3

LexiGrid 5.54.15 - SMTP Configuration and Distributed Workers 3

Purpose and architecture . . . 3
Prerequisites . . . 4
Step 1 - Open the administration page . . . 4
Step 2 - Enter SMTP settings . . . 4
Step 3 - Test connectivity . . . 4
Example: configured SMTP . . . 4
Settings precedence . . . 4
Distributed worker deployment . . . 4
SQLite behavior . . . 4
Scheduled reminders . . . 5
Security boundaries . . . 5
Operational limitations . . . 5
Troubleshooting . . . 5
Upgrade procedure . . . 5
Verification and acceptance criteria . . . 5
Workflow channel recovery . . . 5

Email configuration operations

Global configuration and roles

SMTP configuration is global to the deployment, not per tenant. Platform administrators can edit browser-managed settings; tenant administrators can inspect permitted redacted status and test connectivity. Contact opt-in and queue/recovery operations remain tenant-scoped. SMTP settings never grant application permissions.

Configuration procedure

Open Email Configuration and inspect host, sender, port, security, credential presence, revision and source. Provide approved host/sender and either STARTTLS (commonly 587) or implicit TLS (commonly 465). Enter username/password as required. Leave a saved password blank to retain it; use the explicit clear control to remove it. Submit once and handle stale revision by refreshing rather than overwriting another operator's change.

The password is write-only through the UI and encrypted in the database with the existing LEXIGRID_MAIL_CONFIG_KEY. All API/scheduler/worker instances must retain that same key. A new key is not a repair for unreadable stored overrides. Database settings override environment defaults for subsequent sends; environment/worker-process changes require restart.

Verification without accidental send

Test SMTP connection and authentication performs the configured TLS/authentication checks and NOOP; it does not send an email. Verify actual delivery separately using one consenting designated recipient and an approved test risk. Confirm queue status, SMTP acceptance, any signed provider callback and optional user acknowledgement independently. A green connection test is not proof of inbox delivery.

Error recovery

For ready=false, inspect missing fields and validated mode/port/credential pairing. For failed connections, check DNS, egress, CA trust and provider policy without dumping credentials. For saved override decryption errors, restore access to the original deployment key. For queued messages, inspect worker activation and due/backoff/lease state. Browser editing does not start a stopped worker process.

Production external SMTP/provider interoperability must be qualified independently. Loopback TLS SMTP and crash/retry tests demonstrate the application's acceptance/duplicate boundary, not every provider's behavior.

LexiGrid 5.54.15 - SMTP Configuration and Distributed Workers

Author: Shahryar Jahangir, Aniza Inc.

Purpose and architecture

LexiGrid provides browser-based SMTP administration and concurrent risk notification processing. SMTP settings are global, not tenant-specific. Only platform administrators may change them. Tenant administrators may inspect status and test connectivity.

Prerequisites

A stable 32-byte base64url LEXIGRID_MAIL_CONFIG_KEY must be configured identically on every application instance. Compose passes the existing key and the SMTP environment defaults to both the API and workflow scheduler. Configure LEXIGRID_SMTP_USERNAME and LEXIGRID_SMTP_SECURITY in the Compose environment. The older LEXIGRID_SMTP_USER value remains a fallback for the username. The installer generates the key for new native installations and adds it to existing native configuration files if missing. Back up the key securely. Losing it makes previously saved SMTP overrides unreadable.

Step 1 - Open the administration page

Sign in as platform administrator. Open Email Configuration. The page displays the active SMTP host, port, security mode, sender, configuration revision, and whether credentials are present. Password values are never returned to the browser.

Step 2 - Enter SMTP settings

Enter the SMTP host (for example mail.example.com), port (usually 587 for STARTTLS or 465 for implicit TLS), security mode, sender address, and SMTP username. Enter a new password only when changing it; leave the password blank to retain the current one. Select Clear existing password to remove it. Save configuration.

Step 3 - Test connectivity

Select Test SMTP connection and authentication. The test opens a TLS-protected SMTP connection, authenticates if configured, and performs NOOP. It does not send an email. Send a risk update to a consenting test recipient to validate actual delivery.

Example: configured SMTP

Host: mail.example.com | Port: 587 | Security: STARTTLS | From: lexigrid@example.com | Username: mailer@example.com | Password: entered through the protected form and encrypted at rest. The page does not display the password.

Settings precedence

Browser-managed SMTP settings are encrypted in the shared database and override the corresponding environment settings. Environment settings provide fallback defaults. Changes take effect on the next email send; restarting is not required for SMTP changes. LEXIGRID_MAIL_WORKER is deployment configuration and changing it requires restarting the worker process.

Distributed worker deployment

For multiple worker processes, use a shared PostgreSQL database and set LEXIGRID_MAIL_WORKER=1 on each designated worker instance. The workers claim pending notifications using SELECT FOR UPDATE SKIP LOCKED. A claim is leased for 10 minutes. Other workers can continue processing different messages. Each worker must use the same SMTP encryption key and database.

SQLite behavior

SQLite supports atomic compare-and-swap claiming for local concurrent processes but is not recommended for multi-host deployment. Use PostgreSQL for a production distributed-worker topology. SMTP delivery is at-least-once: a crash after SMTP acceptance but before status update may result in duplicate email.

Scheduled reminders

Reminder records are claimed atomically. Each reminder is advanced to its next due time within the same transaction that queues its risk updates. Closed risks are skipped. Reminder timing depends on a running worker; the worker polls approximately every 60 seconds.

Security boundaries

SMTP credentials are encrypted at rest using AES-256-GCM and a dedicated deployment key. The API never returns the password. Changes require the platform administrator role and are audited. All application instances must use trusted TLS and restricted database credentials. Protect backups and database snapshots; a copy of the encryption key plus database permits decryption.

Operational limitations

The browser cannot activate a stopped worker process: set LEXIGRID_MAIL_WORKER=1 in deployment configuration and restart. This release does not implement distributed leadership election, an external queue, or guaranteed exactly-once email delivery. Provider delivery receipts still require a separately configured signed webhook adapter.

Troubleshooting

SMTP ready=false: inspect validation errors and check host, sender, and credentials. Test connection fails: check DNS, firewall, SMTP TLS mode, provider authentication, and restricted application logs. Worker disabled: enable the worker in deployment config and restart. Key error: confirm the same original LEXIGRID_MAIL_CONFIG_KEY is present on all nodes; never replace an active key without a controlled rotation process.

Upgrade procedure

Back up the database and current .env/.env.native. Upgrade using the installer and preserve database/configuration. After a verified backup/restore drill, apply migrations to the schema required by the installed distribution using the schema-owner connection. Use restricted runtime credentials for the API/workers. Verify the new mail encryption key is present. Restart the application, open Email Configuration, save SMTP settings, test connectivity, and verify a notification from a test risk.

Verification and acceptance criteria

Confirm platform administrator can save SMTP changes; tenant administrator cannot. Confirm the password is absent from API responses and database plaintext. Confirm stale revision is rejected. Confirm two workers process distinct pending messages under PostgreSQL row locking and that reminders do not double-enqueue. Confirm failed SMTP sends retry and appear in notification history.

Workflow channel recovery

Workflow delivery records are tenant-linked and channel-specific. Five automatic attempts use 2/4/8/16-minute backoff and ten-minute reclaimable leases. Stale workers cannot finish a newer claim. Failed channels are listed in Workflows; tenant administrators/risk managers may request one additional attempt with a reason and current attempt count. Platform administrators act only within their selected tenant. A concurrent/stale request returns 409. Attempts remain cumulative, and successful channels are not retried because another channel failed. Stable webhook Idempotency-Key and email Message-ID support receiver deduplication; crash-boundary duplication remains possible.